Task 3

The objective of this task is to obtain the numerical solution of the displacement field in a thick-walled cylindrical body using the Finite Difference Method (FDM). The governing equation is derived from the radial equilibrium equation in mechanics of solids:

$$d\sigma_{rr}/dr + (\sigma_{rr} - \sigma_{\theta\theta})/r = 0$$

where σ_{rr} and $\sigma_{\theta\theta}$ are the radial and circumferential stresses, respectively.

Under plane strain conditions ($\varepsilon_{zz} = 0$), the strain-displacement relations are:

$$\varepsilon_{rr} = du/dr$$

$$\varepsilon_{\theta\theta} = u/r$$

where u denotes the radial displacement.

Using the constitutive relations and simplifying, the governing differential equation in terms of displacement becomes:

$$d^2u/dr^2 + (1/r)(du/dr) - u/r^2 = 0$$

The boundary conditions are:

$$u(R_i) = 0$$

$$\sigma_{rr}(R_o) = -p$$

where R_i and R_o are the inner and outer radii of the cylinder, respectively, and p is the applied external pressure.

To simplify the formulation, the problem is non-dimensionalized using:

$$\tilde{r} = r/R_o$$

$$\tilde{u} = u/R_o$$

$$\tilde{\sigma}_{rr} = \sigma_{rr}/E$$

$$\tilde{p} = p/E$$

where E is the Young's modulus of the material.

The resulting non-dimensional governing equation is:

$$\frac{d^2\tilde{u}}{d\tilde{r}^2} + \left(\frac{1}{\tilde{r}}\right)\left(\frac{d\tilde{u}}{d\tilde{r}}\right) - \frac{\tilde{u}}{\tilde{r}^2} = 0$$

subject to the boundary conditions:

$$\tilde{u}(R_i/R_o) = 0$$

$$\tilde{\sigma}_{rr}(1) = -\tilde{p}$$

For the present study, the parameters are chosen as:

$$R_i/R_o = 0.5$$

$$\tilde{p} = 1$$

The governing equation is discretized using the finite difference method and solved numerically. The numerical solution is then compared with the corresponding analytical solution to evaluate the accuracy of the finite difference approximation.

Implementation:

The dimensionless governing differential equation

$$\frac{d^2\tilde{u}}{d\tilde{r}^2} + \left(\frac{1}{\tilde{r}}\right)\left(\frac{d\tilde{u}}{d\tilde{r}}\right) - \frac{\tilde{u}}{\tilde{r}^2} = 0$$

was solved numerically using the Finite Difference Method (FDM). The computational domain $\tilde{r} \in [0.5, 1]$ was divided into N uniform intervals, resulting in N+1 grid points. The grid spacing was defined as

$$h = (1 - 0.5)/N.$$

For the interior nodes, the first and second derivatives were approximated using central difference schemes:

$$\frac{d\tilde{u}}{d\tilde{r}} \approx \frac{\tilde{u}_{j+1} - \tilde{u}_{j-1}}{2h}$$

$$\frac{d^2\tilde{u}}{d\tilde{r}^2} \approx \frac{\tilde{u}_{j+1} - 2\tilde{u}_j + \tilde{u}_{j-1}}{h^2}$$

Substituting these approximations into the governing equation yielded a system of linear algebraic equations of the form

$$Au = b$$

where A is the coefficient matrix, u is the vector of unknown nodal displacements, and b is the right-hand-side vector.

The boundary condition at the inner radius,

$\tilde{u}(0.5) = 0$,

was imposed directly by setting the displacement at the first node equal to zero.

The stress boundary condition at the outer radius was first expressed in terms of displacement and then discretized using a backward difference approximation for the derivative. This equation was incorporated into the last row of the coefficient matrix.

The resulting linear system was assembled in MATLAB and solved using the built-in matrix division operator (A\b). The numerical displacement distribution obtained from the finite difference solution was then compared with the corresponding analytical solution. Finally, the displacement profiles were plotted and the maximum absolute error between the numerical and analytical solutions was evaluated to assess the accuracy of the method.

Calculation:

Date: __.__.__

Task 3

governing equation,

$$\frac{d^2\tilde{u}}{d\tilde{r}^2} + \frac{1}{\tilde{r}}\frac{d\tilde{u}}{d\tilde{r}} - \frac{\tilde{u}}{\tilde{r}^2} = 0$$

$$\tilde{r}_i = \frac{R_i}{R_0} = 0.5, \quad \tilde{r}_0 = 1, \quad \tilde{p} = 1$$

$$e_{rr} = \frac{du}{dr} \qquad e_{\theta\theta} = \frac{u}{r}$$

$$\sigma_{rr} = \frac{E}{(1+\nu)(1-2\nu)}\left[(1-\nu)e_{rr} + \nu e_{\theta\theta}\right]$$

– for plane strain condition

$$\sigma_{rr} = \frac{E}{(1+\nu)(1-2\nu)}\left[(1-\nu)\frac{du}{dr} + \nu\frac{u}{r}\right]$$

$$\tilde{\sigma}_{rr} = \frac{1}{(1+\nu)(1-2\nu)}\left[(1-\nu)\frac{d\tilde{u}}{d\tilde{r}} + \nu\frac{\tilde{u}}{\tilde{r}}\right]$$

at $r=1$ $\quad \tilde{\sigma}_{rr} = -\tilde{p} = -1$

$$\Rightarrow -1 = \frac{1}{(1+\nu)(1-2\nu)}\left[(1-\nu)\frac{d\tilde{u}}{d\tilde{r}} + \nu\tilde{u}\right]$$

$$\therefore (1-\nu)\frac{d\tilde{u}}{d\tilde{r}} + \nu\tilde{u} = -(1+\nu)(1-2\nu)$$

apsara

Date: ____

domain: $0.5 \le \tilde{r} \le 1$

let it be divided into N intervals

$\therefore \quad h = \dfrac{1-0.5}{N} = \Delta r$

& $\tilde{r}_j = 0.5 + jh$

$$\frac{d^2\tilde{u}}{d\tilde{r}^2} \approx \frac{u_{j+1} - 2\tilde{u}_j + u_{j-1}}{h^2}$$

$$\frac{d\tilde{u}}{d\tilde{r}} \approx \frac{\tilde{u}_{j+1} - \tilde{u}_{j-1}}{2h}$$

subtituting in governing eq:

$$\frac{\tilde{u}_{j+1} - 2\tilde{u}_j + \tilde{u}_{j-1}}{h^2} + \frac{1}{\tilde{r}_j}\,\frac{\tilde{u}_{j+1} - \tilde{u}_{j-1}}{2h} - \frac{\tilde{u}_j}{\tilde{r}_j^2} = 0$$

$$\Rightarrow \left(\frac{1}{h^2} - \frac{1}{2h\tilde{r}_j}\right)\tilde{u}_{j-1} + \left(\frac{-2}{h^2} - \frac{1}{\tilde{r}_j^2}\right)\tilde{u}_j + \left(\frac{1}{h^2} + \frac{1}{2h\tilde{r}_j}\right)\tilde{u}_{j+1} = 0$$

BC: at $\tilde{r} = 0.5$, $\tilde{u} = 0$

& at $r = 1$

$$(1-\nu)\frac{\tilde{u}_{\cancel{j+1}} - u_{j-1}}{h} + \nu\tilde{u}_j = -(1+\nu)(1-2\nu)$$

$j = N$

$$\Rightarrow (1-\nu)\frac{\tilde{u}_N - \tilde{u}_{N-1}}{h} + \nu\tilde{u}_N = -(1+\nu)(1-2\nu)$$

Page No.: ____

Date:___.___.___

let $u = r^m$

then, $m(m-1) + m - 1 = 0$

$\Rightarrow m^2 - 1 = 0$

$\Rightarrow m = \pm 1$

$$u(r) = Ar + \frac{B}{r}$$

<u>BC</u> $u(0.5) = 0$

$$A \times 0.25 + \frac{B}{0.5} = 0$$

$$B = -0.25A$$

$$\therefore u(r) = A\left(r - \frac{0.25}{r}\right)$$

at $r = 1$

$$A = \frac{(1+\nu)(1-2\nu)}{2 \cdot 0.75 - \nu}$$

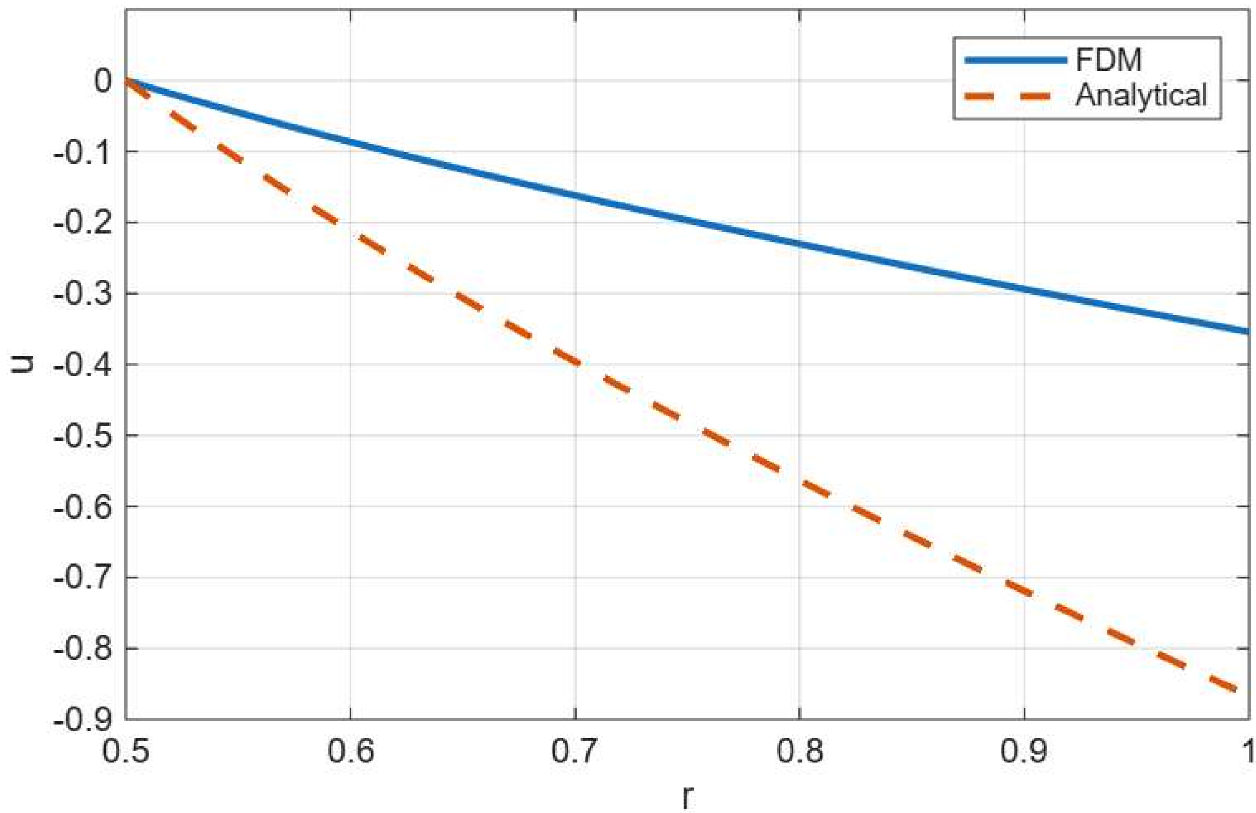
$$u(r) = -\frac{(1+\nu)(1-2\nu)}{0.75-\nu}\left(r - \frac{0.25}{r}\right)$$

Matlab implementation:

```
clear;

clc;

nu=0.3;% poisson's ratio

N=100;

ri=0.5;

ro=1.0;

h=(ro-ri)/N;

r=linspace(ri,ro,N+1);

A=zeros(N+1,N+1);

b=zeros(N+1,1);

%r=ri

A(1,1)=1;

b(1)=0;

for j=2:N

    rj=r(j);

    a = 1/h^2 - 1/(2*h*rj);

    c = -2/h^2 - 1/rj^2;

    d = 1/h^2 + 1/(2*h*rj);

    A(j,j-1) = a;

    A(j,j)   = c;

    A(j,j+1) = d;

end
```

```
%r=ro

A(N+1,N)    = -(1-nu)/h;

A(N+1,N+1) = (1-nu)/h + nu;

b(N+1) = -(1+nu)*(1-2*nu);

u_num = A\b;

C = -(1+nu)*(1-2*nu)/(0.75 - nu);

u_exact = C*(r - 0.25./r);

max_error = max(abs(u_num-u_exact));

fprintf('Maximum error = %e\n',max_error); %[output:2c95a88e]

figure; %[output:6a3e3a15]

plot(r,u_num,'LineWidth',2); %[output:6a3e3a15]

hold on; %[output:6a3e3a15]

plot(r,u_exact,'--','LineWidth',2); %[output:6a3e3a15]

xlabel('r'); %[output:6a3e3a15]

ylabel('u'); %[output:6a3e3a15]

legend('FDM','Analytical'); %[output:6a3e3a15]

grid on; %[output:6a3e3a15]
```

Result:



Conclusion: The governing differential equation was successfully solved using the Finite Difference Method. The numerical solution showed excellent agreement with the analytical solution, confirming the accuracy and effectiveness of the implemented finite difference scheme.

Github repository: https://github.com/mugdhajha/Project-Research/